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Catalysts and inhibitors of critical transitions in ecological systems

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Ecological systems can experience sudden and often irreversible regime shifts, also known as critical transitions, with major consequences such as desertification, locust outbreaks, and coral reef collapse. Anticipating these shifts is a central challenge, particularly under accelerating climate change. Although early warning signals of critical transitions have been widely studied, the mechanisms that drive or prevent them remain less well understood. Here, we develop a theoretical framework based on time-delayed dynamics that allows us to identify processes acting as catalysts or inhibitors of critical transitions in ecological systems. We show that a composite measure combining time-delayed species interactions with species abundances is a key modulator of critical transitions. Beyond the critical point, systems exhibit persistent abundance oscillations, substantially increasing the risk of large-scale destabilization and species extinctions. Additionally, we show that a high diversity of species interaction types can act as a buffer of critical transitions. Instead, strong species self-regulation effects can act as catalysts of such transitions, contrary to common expectations. We illustrate the framework with empirical data from microbial systems. Together, these results provide a formal platform for exploring and understanding the drivers of critical transitions in complex living systems.

complex systems | critical transitions | interactions | time delays

Significance

Critical transitions—sudden shifts in the state of ecological systems—can trigger phenomena such as desertification, locust outbreaks, and coral reef collapse, with far-reaching ecological and societal consequences. While early warning signals of such transitions have been studied extensively, the underlying drivers that accelerate or suppress these transitions remain less clear. We develop a theoretical framework based on time-delayed dynamics that helps disentangle the roles of different processes as catalysts or inhibitors of critical transitions. We illustrate the framework with empirical data from microbial systems. This approach provides a foundation for systematically investigating the mechanisms that govern critical transitions in complex living systems.

Critical transitions—sudden and often irreversible shifts in ecological states that occur when environmental drivers cross threshold values—are increasingly recognized as fundamental features of complex ecosystems (1–5). Such critical transitions can push systems into alternative stable states with major ecological and societal consequences, including desertification (6), locust outbreaks (7), and coral reef degradation (8). Anticipating these transitions has become an urgent goal in the context of accelerating climate change (4, 9), motivating extensive theoretical and empirical work on their dynamics (3, 5, 10–26). In particular, research on early warning signals has advanced rapidly, with critical slowing down indicators providing important tools for anticipating the approach of critical transitions (3–5, 10, 27–29). For a valuable overview on critical slowing down indicators, see the perspective (27). Other studies have examined the mechanisms underlying alternative stable states (11–13) and the conditions that allow large ecological networks to maintain stability or lose it (16–26). Despite these advances, however, a central open question remains: What factors act as catalysts that accelerate critical transitions, and which processes function as inhibitors that delay or prevent them? Addressing this gap is essential for developing timely and effective strategies to mitigate undesirable ecosystem shifts.

A common assumption in most studies of ecological dynamics is that species respond to external changes instantaneously. Yet this simplification may obscure our ability to determine which processes act as catalysts or inhibitors of critical transitions. In natural systems, many responses are delayed because biological and ecological processes take time—for example, maturation periods (30, 31), behavioral reaction times (30, 32), or resource regeneration lags (30, 33, 34). A classic case is predator–prey dynamics, where predators must process and convert consumed prey before population growth occurs, producing a delay between prey fluctuations and predator responses. Because such time delays can arise from multiple mechanisms, they are likely widespread in ecological systems (30, 34, 35) and may play an important role in shaping when and how critical points are reached.

Previous work has highlighted the importance of time delays in shaping ecological dynamics. Classic analyses of one- and two-species systems demonstrated that even

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modest lags—such as those associated with growth or reproduction—can destabilize systems that would otherwise remain stable (30, 33–36). More recent studies of species-rich systems with time delays have shown that delays can modify established stability principles (37–39). In particular, they indicate that classical stability bounds may not hold; short delays can promote stability, whereas longer delays tend to undermine it; and communities with a mix of interaction types often exhibit higher levels of stability. With respect to critical transitions, these studies also suggest that increasing either time delays or equilibrium abundances toward threshold values can induce instability and precipitate a transition. Yet most of these studies have focused on systems with randomly assigned interactions (37–39), whereas real ecological communities often deviate markedly from randomness, exhibiting structured networks of competitive, mutualistic, and exploitative interactions. Because such structures pose substantial mathematical challenges, a comprehensive understanding of critical transitions in ecologically realistic systems remains limited.

Motivated by this gap, we develop a framework to systematically investigate the catalysts and inhibitors of critical transitions in complex ecological systems with time-delayed interactions at equilibrium points. Our analyses identify a unifying quantity that links species abundances with time-delayed interactions and provide a closed-form theoretical expression for the location of critical points in a broad class of systems with random interaction coefficients. This formulation allows us to examine how global system properties shape the conditions under which critical transitions occur, and in turn to assess which factors can act as catalysts or inhibitors. We illustrate the applicability of this approach using empirical data from microbial systems. More broadly, we view our study as a step toward clarifying the drivers of critical points in complex ecological dynamics.

Results

Modeling Framework. We represent an ecological community of S interacting species using a continuous-time dynamical system with delayed interactions:

$$\frac{d\mathbf{X}(t)}{dt} = \mathrm{diag}(\mathbf{X}(t))\ \mathbf{f}(\mathbf{X}(t-\tau))\,. \qquad [1]$$

Here, $\mathbf{X}(t) = \big(X_1(t), X_2(t), \ldots, X_S(t)\big) \in \mathbb{R}^S$ denotes the vector of species abundances at time t, and $\mathrm{diag}(\mathbf{X}(t))$ is the diagonal matrix with entries $X_i(t)$ on its diagonal. The function $\mathbf{f}(\mathbf{X}(t-\tau))$ encodes the network of species interactions, evaluated at a delayed state of the system. The parameter τ captures the fact that species typically do not respond to changes in their own or others' abundances instantaneously, but only after a characteristic lag arising from processes such as maturation, reaction times, or resource regeneration.

We focus on point equilibria $\mathbf{X}^*$, where $X_i^* > 0$ for all i, meaning that every species persists and the equilibrium is therefore feasible (40–42). By Eq. **1**, such equilibria satisfy $\mathbf{f}(\mathbf{X}^*) = \mathbf{0}$. The dynamics near a feasible coexistence equilibrium $\mathbf{X}^*$ can be captured by the linearized system

$$\frac{d\mathbf{x}(t)}{dt} = \mathrm{diag}(\mathbf{X}^*)\ \mathbf{J}\,\mathbf{x}(t-\tau) = \mathbf{M}\,\mathbf{x}(t-\tau), \qquad [2]$$

where $\mathbf{x}(t) = \mathbf{X}(t) - \mathbf{X}^*$ denotes deviations from equilibrium. The element J_{ij} of the matrix $\mathbf{J}$ quantifies the per capita effect of species j on species i in the vicinity of $\mathbf{X}^*$, with the diagonal entries representing self-regulation of each species. We refer to $\mathbf{J}$ as the interaction matrix, and define the community matrix as $\mathbf{M} = \mathrm{diag}(\mathbf{X}^*)\,\mathbf{J}$ (17, 34, 43–45).

According to bifurcation theory (30, 46), the conditions for a critical transition can be analyzed through the roots of the associated characteristic equation:

$$H(z) = z - \lambda\, e^{-z\tau}, \qquad [3]$$

as derived in *SI Appendix*, section 1. Here, λ denotes an eigenvalue of the community matrix $\mathbf{M}$. A transition occurs when the real part of the dominant characteristic root z_1—the root with the largest real component—crosses zero in the complex plane, signaling a loss of stability of the equilibrium state.

Type and Driver of Critical Transitions in Time-Delayed Systems. As a first step, we introduce two simplifying assumptions: all species share the same time lag τ in their responses, and they coexist at a common equilibrium abundance X^* (these assumptions are relaxed later). Within such time-delayed systems governed by a fixed interaction matrix, critical transitions can arise through two pathways: either by increasing the time lag τ or by increasing the equilibrium abundance X^*.

To illustrate these mechanisms, we consider a single-species case based on the classic logistic model with a time delay (36) (Fig. 1*A*). In this system, the delay generates a mismatch between the current abundance of the species and the growth rate, which depends on its abundance at an earlier time $t - \tau$. External perturbations can amplify this mismatch, producing oscillations in abundance. When the time delay or equilibrium abundance remains small, the mismatch is limited, the oscillations are damped, and the system eventually returns to its original equilibrium (Fig. 1 *B* and *C*, *Left* panels). In this regime, the equilibrium remains stable and no critical transition occurs (Fig. 1 *D* and *E*, purple regions).

Increasing the time delay or the equilibrium abundance beyond a critical value produces a strong mismatch between current abundances and delayed responses. In this regime, perturbation-induced oscillations no longer dissipate; instead, the system settles into persistent oscillations (Fig. 1 *B* and *C*, *Right* panels). This shift from a stable fixed point to sustained oscillatory dynamics marks the occurrence of a critical transition (Fig. 1 *D* and *E*). The resulting oscillations cause continuous fluctuations in species abundances, which may substantially increase the risk of large-scale extinctions (Fig. 1 and *SI Appendix*, Fig. S1). It is worth noting that entering such oscillatory regimes does not necessarily mean that the system loses resilience to perturbations, since if the oscillations correspond to a globally stable attractor, the system can, in principle, withstand arbitrary perturbations and return to the oscillating attractor. Mathematically, this transition corresponds to a Hopf bifurcation (*SI Appendix*, section 1 and Fig. S2). Extending the analysis to species-rich systems with different interaction types shows that critical transitions driven by either time delays or equilibrium abundances also take the form of Hopf bifurcations (*SI Appendix*, Figs. S3–S6).

Since both time delay and equilibrium abundance can independently trigger critical transitions in delayed systems, a natural question is whether a single parameter can capture their combined effect. This can be addressed by analyzing the characteristic equation (Eq. **3**). Substituting the relationship between the eigenvalues of the community matrix and the interaction matrix, $\lambda = X^*\lambda_{\mathbf{J}}$, shows that the stability of the equilibrium depends

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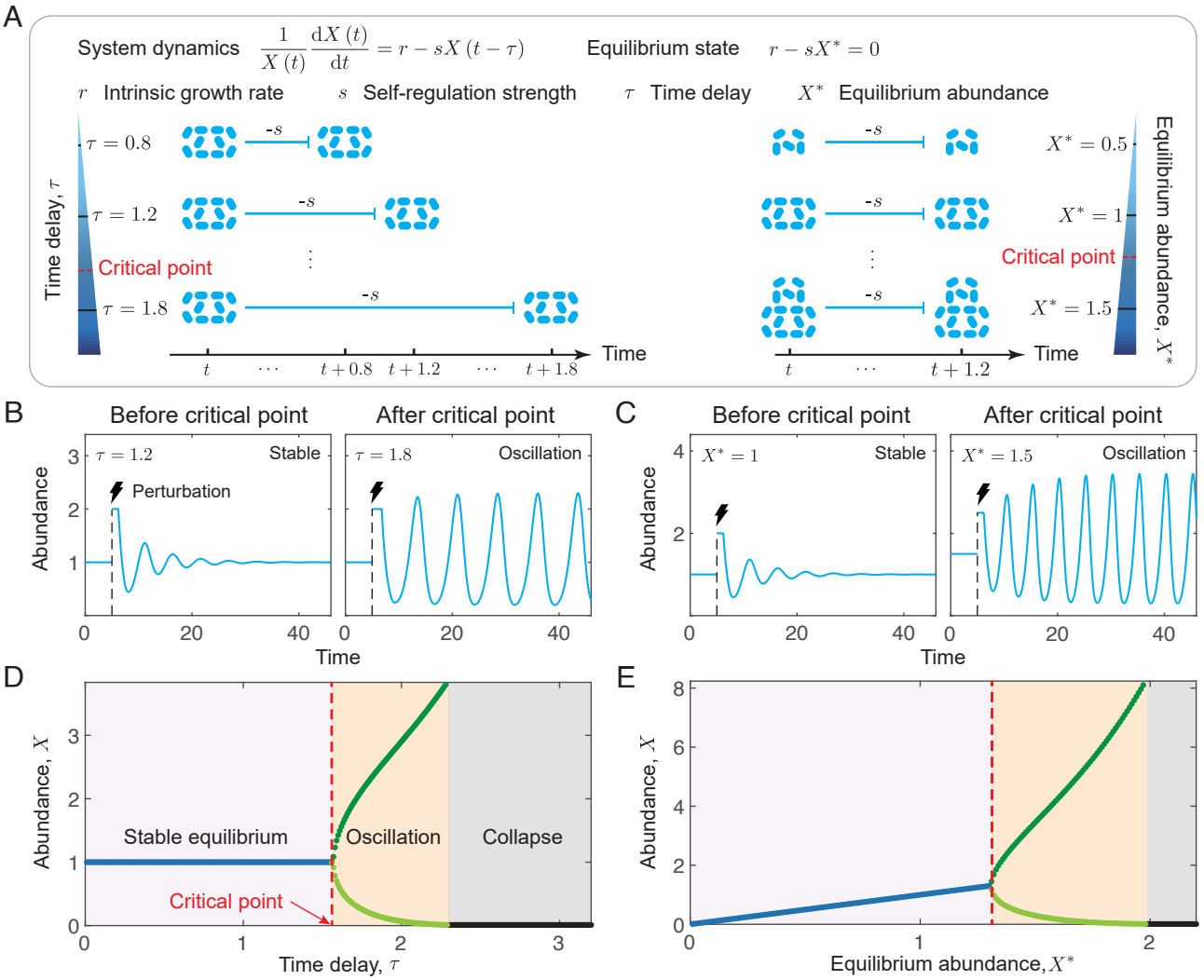


Fig. 1. Critical transitions in time-delayed systems. (*A*) Schematic of a single-species system with different combinations of time delay τ and equilibrium abundance X^*. Blue rods represent the population, with the number of rods indicating the equilibrium abundance. Blunt-head arrows denote negative self-regulation, pointing from the population at time t to its state at a later time to illustrate delayed feedback. Populations in the left half share $X^* = 1$ but differ in τ, whereas those in the right half share $\tau = 1.2$ but differ in X^*. Increasing either time delay or equilibrium abundance beyond a threshold (the critical point; dashed red line) induces a critical transition, which is reflected in system responses to perturbations (*B* and *C*). (*B* and *C*) Dynamical trajectories under external perturbations. Before the critical point (*Left* panels), oscillations are damped and the system returns to equilibrium. After the critical point (*Right* panels), oscillations persist and the equilibrium loses stability, indicating that a critical transition has occurred. (*D* and *E*) Bifurcation diagrams of the single-species system with respect to τ (*D*) and X^* (*E*). When the critical point (red dashed line) is not reached (purple region), the equilibrium (blue dots) remains stable. Once the critical point is crossed (orange region), the system undergoes a Hopf bifurcation and enters an oscillatory regime. Light and dark green points indicate the minimum and maximum abundances attained during oscillations. Further increases in τ or X^* eventually lead to system collapse (gray region). Overall, the figure shows that exceeding a critical threshold in time delay or equilibrium abundance drives the system from stability to oscillations and eventually collapse. Numerical simulations in panels *B*–*E* are based on the logistic model with $s = 1$.

on the product τX^*. To maintain stability, all characteristic roots z must have negative real parts, which is equivalent to requiring that every eigenvalue of $\mathbf{J}$ lies within a teardrop-shaped region of the complex plane defined by

$$\begin{cases} \tau X^* = \dfrac{1}{\sqrt{x^2+y^2}} \tan^{-1}\left(-\dfrac{x}{y}\right), & x \le 0,\ y \ge 0, \\ \tau X^* = \dfrac{1}{\sqrt{x^2+y^2}} \tan^{-1}\left(\dfrac{x}{y}\right), & x \le 0,\ y < 0, \end{cases} \quad [4]$$

as derived in *SI Appendix*, section 1 and Fig. S7. Here, x and y denote the real and imaginary components of the eigenvalues, respectively. As τX^* increases, the stability region contracts, making it more likely that eigenvalues of $\mathbf{J}$ cross its boundary. Crossing this boundary corresponds to the onset of a critical transition. Thus, the product τX^* emerges as the effective bifurcation parameter governing the loss of stability in these systems.

Numerical simulations confirm this conclusion, showing that τ and X^* vary inversely at the critical points (Fig. 2). Motivated by this observation, we define the product $\tau_{\text{eff}} = \tau X^*$ as the effective delay, which serves as the bifurcation parameter driving critical transitions in time-delayed systems. We denote by τ^*_{eff} the value of the effective delay at which a critical point is reached. Systems with larger τ^*_{eff} therefore tend to remain farther from critical transitions under comparable conditions. In the following sections, we use this notion of the critical point to examine how fundamental system properties shape the likelihood of such transitions.

Estimating the Critical Point of Time-Delayed Systems. Having identified the effective delay τ_{eff} as the driver of critical transitions, we next analyze the corresponding critical points across systems

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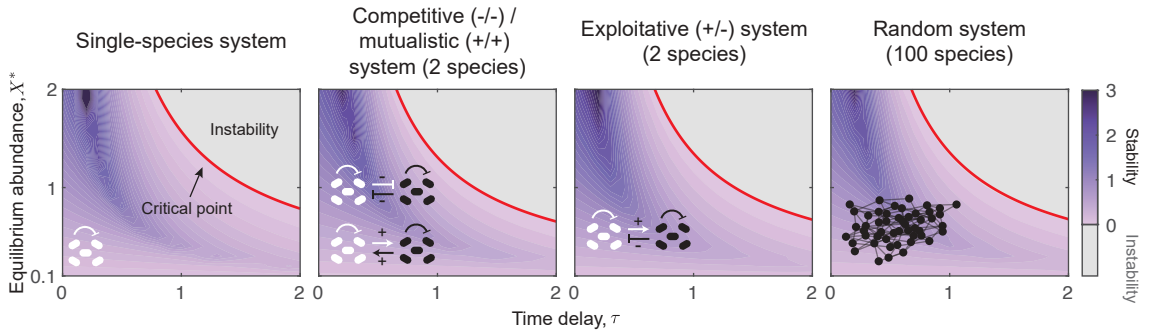


Fig. 2. Phase diagrams of systems with different interaction types. *Left* three panels: Black and white rods represent individuals of different species. Blunt-head arrows denote negative effects (competition or self-regulation), and sharp-head arrows denote positive effects (facilitation). *Rightmost* panel: Black dots represent species, with connecting lines indicating interspecific interactions. In each panel, the purple and gray regions, obtained numerically, correspond to stable and unstable system phases, respectively, as indicated by the color bar. The red line marks the boundary (critical point) between these regions. When the combination of equilibrium abundance X^* and time delay τ lies within the purple region, the system remains stable and no critical transition occurs. When the combination falls within the gray region, the equilibrium loses stability and the system undergoes a critical transition. Across all cases, this boundary follows an inverse proportional relationship between X^* and τ. These results highlight that the product of equilibrium species abundance and time delay drives the critical transition in time-delayed systems. For all systems, we set $s = 1$. Additional parameters for the two-species systems are $J_{12} = J_{21} = 0.2$ (mutualistic), $J_{12} = J_{21} = -0.2$ (competitive), and $J_{12} = 0.2, J_{21} = -0.2$ (exploitative). For the random system, parameters are $S = 100$, $C = 0.1$, and $\sigma = 0.05$.

of increasing complexity, beginning with single- and two-species models before turning to larger communities with diverse interaction types.

For a single-species system, the interaction matrix $\mathbf{J}$ has only one eigenvalue, $\lambda_{\mathbf{J},1} = -s$. At the critical point, this eigenvalue approaches the left endpoint of the teardrop-shaped stability region, yielding the condition

$$\tau^*_{\text{eff}} = \frac{\pi}{2s}, \qquad [5]$$

as shown in *SI Appendix*, Fig. S8 and derived in *SI Appendix*, section 2 (33, 35, 36).

For a two-species system, the interaction matrix $\mathbf{J}$ has two eigenvalues: $\lambda_{\mathbf{J},1} = -s + \sqrt{J_{12}J_{21}}$ and $\lambda_{\mathbf{J},2} = -s - \sqrt{J_{12}J_{21}}$. When the interaction is competitive (-/-) or mutualistic (+/+), we have $J_{12}J_{21} > 0$, and both eigenvalues are real. In this case, a critical transition occurs when $\lambda_{\mathbf{J},2}$ approaches the left endpoint of the teardrop-shaped stability region, leading to the critical point

$$\tau^*_{\text{eff}} = \frac{\pi}{2\left(s + \sqrt{J_{12}J_{21}}\right)}. \qquad [6]$$

(*SI Appendix*, Fig. S8 and section 2).

By contrast, if the interaction is exploitative (+/-), then $J_{12}J_{21} < 0$, and both eigenvalues are complex. In this case, the critical point is reached when the pair of complex conjugate eigenvalues touches the boundary of the stability region, yielding

$$\tau^*_{\text{eff}} = \frac{\pi}{2\sqrt{s^2 + |J_{12}J_{21}|}} - \frac{\tan^{-1}(|J_{12}J_{21}|/s)}{\sqrt{s^2 + |J_{12}J_{21}|}}. \qquad [7]$$

(*SI Appendix*, Fig. S8 and section 2).

Building on random matrix approaches to community stability (16–19, 21, 22, 25, 26, 38, 39, 47), we extend the analysis to species-rich systems in which interactions may be exploitative (+/-), competitive (-/-), or mutualistic (+/+). In such systems, the eigenvalues of the interaction matrix $\mathbf{J}$ are typically distributed within an ellipse in the complex plane, sometimes accompanied by an isolated outlier (17, 19, 20, 25, 48, 49). Because the matrix entries are real, this distribution is symmetric about the real axis. As the effective delay τ_{eff} increases, the eigenvalues at the edges of this ellipse—particularly the leftmost and the uppermost (or equivalently, lowermost) points—are the most likely to cross the boundary of the teardrop-shaped stability region (*SI Appendix*, Fig. S10). Consequently, these extremal eigenvalues provide a tractable basis for estimating the critical point of species-rich systems.

In systems where interactions are assigned at random, we assume that any pair of species i and j interacts with probability C, and that the per capita effects J_{ij} and J_{ji} are independent realizations of a random variable Z with mean 0 and variance σ^2. In this case, the eigenvalues of $\mathbf{J}$ follow a circular distribution (16, 17, 19), centered at $(-s, 0)$ with radius $\sigma\sqrt{SC}$. For sufficiently large matrices, the critical point is therefore estimated as

$$\tau^*_{\text{eff,random}} = \frac{\pi}{2\sqrt{s^2 + \sigma^2 SC}} - \frac{\tan^{-1}\left(\sigma\sqrt{SC}/s\right)}{\sqrt{s^2 + \sigma^2 SC}}, \qquad [8]$$

as shown in Fig. 3*A* and derived in *SI Appendix*, section 2.

For species-rich systems dominated by exploitative interactions (e.g., predator–prey dynamics), each pair of species interacts with probability C, but the effects have opposite signs: J_{ij} is drawn from $|Z|$, and J_{ji} from $-|Z|$. In this case, the eigenvalues of $\mathbf{J}$ are distributed within a vertically elongated ellipse (17, 19, 48), centered at $(-s, 0)$, with horizontal semiaxis $r_{x,+/-} = \sigma\sqrt{SC}\left(1 - \mathbb{E}^2(|Z|)/\sigma^2\right)$ and vertical semiaxis $r_{y,+/-} = \sigma\sqrt{SC}\left(1 + \mathbb{E}^2(|Z|)/\sigma^2\right)$. For large matrices, the critical point can then be estimated as

$$\tau^*_{\text{eff},+/-} = \frac{\pi}{2\sqrt{s^2 + r^2_{y,+/-}}} - \frac{\tan^{-1}\left(r_{y,+/-}/s\right)}{\sqrt{s^2 + r^2_{y,+/-}}} \qquad [9]$$

(Fig. 3*A* and *SI Appendix*, section 2).

In purely competitive or purely mutualistic systems, each pair of species i and j interacts with probability C, but the interaction strengths share the same sign: both J_{ij} and J_{ji} are drawn from $-|Z|$ for competition, and from $|Z|$ for mutualism. In both cases, the eigenvalues of $\mathbf{J}$ lie within a horizontally elongated ellipse, with a single outlier eigenvalue located outside this bulk distribution (17, 19).

For large competitive systems, the outlier lies on the left side of the ellipse and is therefore most likely to cross the boundary of

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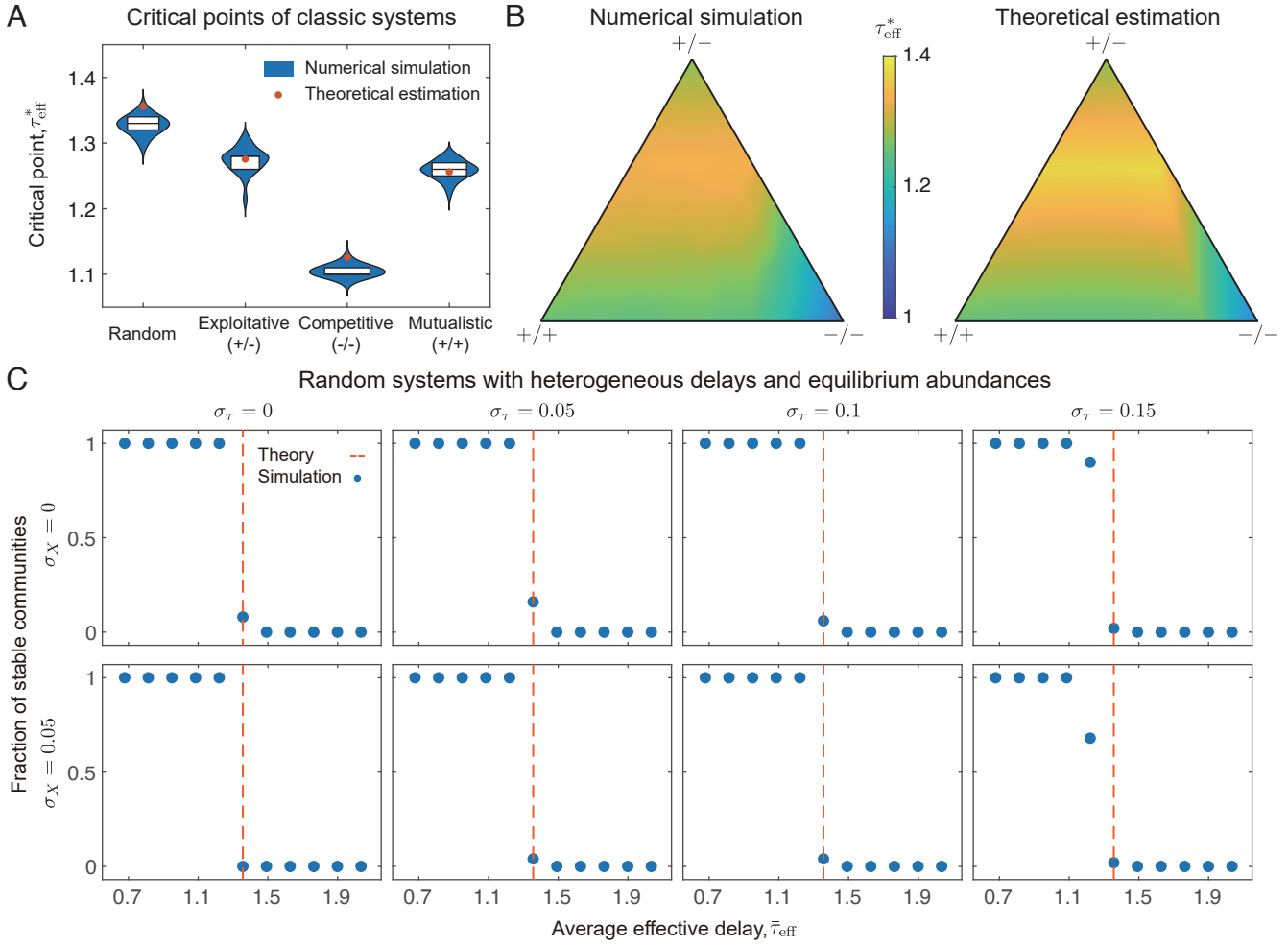


Fig. 3. Estimating the critical point of large systems. (*A*) The critical point $\tau^*_{\text{eff}} = \tau X^*$ for four system types: fully random, exploitative, competitive, and mutualistic. Red dots indicate theoretical estimates, while violin plots show the distribution of numerical simulation results. Each plot is based on 50 independently generated systems with interaction coefficients drawn from the same normal distribution. (*B*) Critical point estimates for systems in which different interaction types are mixed at varying proportions. Colors closer to yellow indicate higher critical points, while colors closer to blue indicate lower values (color bar). The three corners of the triangle correspond to purely mutualistic (+/+), purely exploitative (+/-), and purely competitive (-/-) interactions, while interior points represent mixtures of these interaction types at varying proportions. The *Left* panel shows simulation results (each data point is the average over 30 generated systems), and the *Right* panel shows the corresponding theoretical estimates. These results demonstrate that systems with more diverse interaction types (toward the center of the triangle) tend to exhibit higher critical points, and thus are less likely to approach a critical transition under comparable conditions. (*C*) Critical transitions in systems with heterogeneous delays and equilibrium abundances. In the random systems shown here, species equilibrium abundances are sampled from a uniform distribution with mean $\mu_X = 1$ and variance σ_X^2, while interaction delays are sampled from a uniform distribution with mean μ_τ and variance σ_τ^2. The average effective delay is defined as $\bar{\tau}_{\text{eff}} = \mu_\tau \mu_X$. Orange dashed lines represent theoretical predictions, and blue dots denote simulation results (each based on 50 generated systems with interaction coefficients drawn from the same normal distribution). Simulations follow the Lotka–Volterra model with heterogeneous delays and equilibrium abundances, as described in *Materials and Methods*. Taken together, these results indicate that theoretical predictions of the critical point are in close agreement with numerical simulations, and that systems with diverse interaction types tend to exhibit higher critical points. Parameters for all panels: $S = 100$, $C = 0.1$, $\sigma = 0.05$, and $s = 1$.

the teardrop-shaped stability region. The corresponding critical point is estimated as

$$\tau^*_{\text{eff,-/-}} = \frac{\pi}{2\left(s + (S-1)C\,\mathrm{E}(|Z|)\right)}. \qquad [10]$$

In contrast, for large mutualistic systems the outlier lies on the right side of the ellipse. In this case, the leftmost edge of the ellipse is the first to reach the stability boundary, leading to the critical point

$$\tau^*_{\text{eff,+/+}} = \frac{\pi}{2\left(s + C\,\mathrm{E}(|Z|) + r_{x,+/+}\right)}, \qquad [11]$$

where the horizontal semiaxis of the ellipse is given by

$$r_{x,+/+} = \frac{\sqrt{SC\left(\sigma^2 - C\,\mathrm{E}^2(|Z|)\right)\left(\sigma^2 + (1-2C)\,\mathrm{E}^2(|Z|)\right)}}{\sigma^2 - C\,\mathrm{E}^2(|Z|)}. \qquad [12]$$

(Fig. 3*A* and *SI Appendix*, section 2).

Beyond these four simple interaction types, our theoretical estimation can also be extended to systems in which different interaction types—exploitative, competitive, and mutualistic—occur in arbitrary proportions. This extension provides additional generality and enables us to assess how varying mixtures of interactions influence the location of the critical point (Fig. 3*B*; *Materials and Methods* and *SI Appendix*, section 2).

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As noted at the outset, our analytical results rely on two simplifying assumptions: that all species share the same time delay τ, and that equilibrium is homogeneous, with $X_i^* \equiv X^*$ for every species. These assumptions are unlikely to hold in real ecosystems, but they make the analysis tractable. To evaluate robustness, we relaxed both assumptions numerically and found that the main results persist even under relatively large (∼15%) deviations from perfect equality (Fig. 3*C*; *Materials and Methods*; *SI Appendix*, Figs. S11–S14). We also demonstrate that when variation in equilibrium abundances becomes more pronounced, our theory tends to overestimate the critical points (*SI Appendix*, Fig. S15).

Diversity of Interaction Types Can Inhibit the Emergence of Critical Transitions. How species interact is a central determinant of ecological dynamics. Having derived theoretical expressions for the critical points of different system types, we can now compare how interaction structure influences critical transitions in time-delayed systems. Across the four canonical cases—random, exploitative, competitive, and mutualistic—we find that fully random systems (Eq. **8**) exhibit the highest critical point (Fig. 3*A*). In practice, this implies that for the same system parameters, random systems are the least susceptible to crossing a critical point. This outcome stands in marked contrast to delay-free systems, where exploitative interactions tend to confer the greatest stability against potential losses (17).

This pattern can be understood by examining the eigenvalue distributions of different system types. As noted above, the distributions for exploitative, competitive, and mutualistic systems are distorted relative to the circular distribution of random systems. These distortions make the eigenvalues more likely to exit the teardrop-shaped stability region as the effective delay increases, which helps explain why random systems are less prone to reaching a critical point than the other three cases. Extending the analysis to systems with mixed-interaction types further generalizes this result: the critical point of a system that combines multiple interaction types is higher than that of a system restricted to a single type (Fig. 3*B*). This finding suggests that diversity in interaction types can act as a stabilizing influence in time-delayed systems, potentially helping to account for the prevalence of diverse interaction structures in natural communities (50).

Complexity as a Catalyst of Critical Transitions. The relationship between ecological complexity and system stability has long been debated (16, 51). We approach this question from the perspective of delay-associated critical transitions. While our analysis confirms the stabilizing role of diverse interaction types, it also shows that increasing the quantity $\sigma\sqrt{SC}$—often used as a proxy for ecological "complexity" (16, 17)—has the opposite effect. Larger values of species richness (S), connectance (C), or interaction strength variance (σ^2) expand the eigenvalue distributions of all system types, thereby reducing the effective delay τ_{eff} at which the critical point is reached (Fig. 4). In this sense, higher complexity can act as a catalyst of critical transitions. This result is consistent with May's original analysis of random matrices, which showed that sufficiently diverse and strongly interacting communities are unlikely to remain locally stable. At the same time, the usual caveats of May's result continue to apply, including the focus on local stability around equilibria and the omission of species-specific biological details.

Strong Self-Regulation as a Catalyst of Critical Transitions. Originating from sources such as intraspecific interference, cannibalism, and spatial heterogeneity, self-regulation introduces

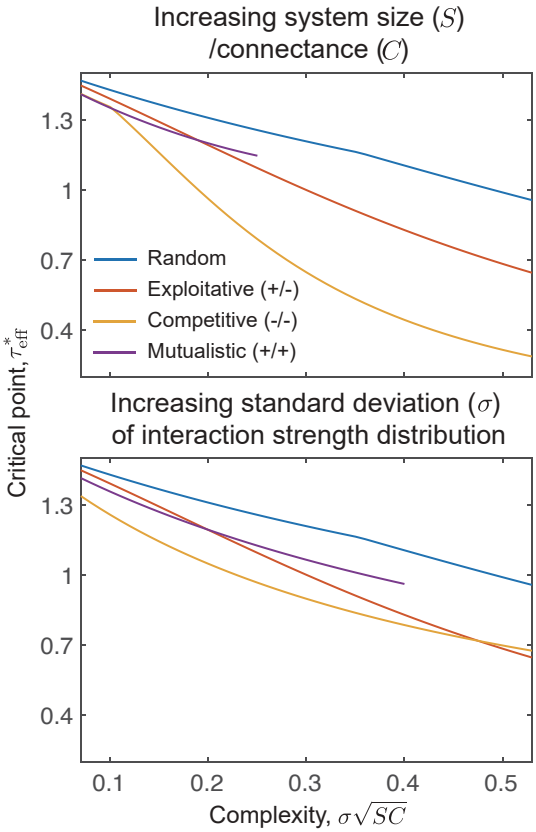


Fig. 4. Relationship between system complexity and the critical point. System complexity is quantified by the "complexity parameter" $\sigma\sqrt{SC}$, where S denotes system size, C the connectance, and σ the SD of the interaction strength distribution. Solid lines indicate theoretical estimates. *Top* panel: Results obtained by increasing system size (S) or connectance (C). *Bottom* panel: Results obtained by increasing the SD σ of the interaction strength distribution. These results suggest that increasing system complexity lowers the critical point, indicating that complexity acts as a catalyst for critical transitions. For cases where S is varied, we set $C = 0.1$ and $\sigma = 0.05$. For cases where C is varied, we set $S = 100$ and $\sigma = 0.05$. For cases where σ is varied, we set $S = 100$ and $C = 0.1$. In all cases, $s = 1$.

a direct negative dependence between per capita growth rate and a species' own abundance—reducing growth rates when abundance is high and increasing them when abundance is low—appearing as negative diagonal terms in the interaction matrix. Self-regulation can therefore be regarded as an important stabilizing mechanism. In delay-free systems, there is ongoing debate over whether strong and ubiquitous self-regulation is truly necessary to achieve stability (52). Nevertheless, existing analyses show that uniformly increasing the strength of self-regulation will, beyond some threshold, stabilize any system and push it farther from the critical point (17, 25). We now ask whether the same conclusion applies to time-delayed systems.

Interestingly, we find that the effect of stronger self-regulation depends on the type of interspecific interactions present (Fig. 5). In systems with few exploitative interactions, increasing self-regulation strength consistently lowers the critical point, making critical transitions more likely (Fig. 5 *B* and *C*). By contrast, when exploitative interactions are common, the relationship becomes nonmonotonic: increasing self-regulation initially raises the critical point and delays the transition, but further increases eventually lower the critical point, thereby promoting the transition (Fig. 5 *B* and *C* and *SI Appendix*, section 3). In other words, the stabilizing role of self-regulation in delay-free systems can be reversed under time-delayed dynamics.

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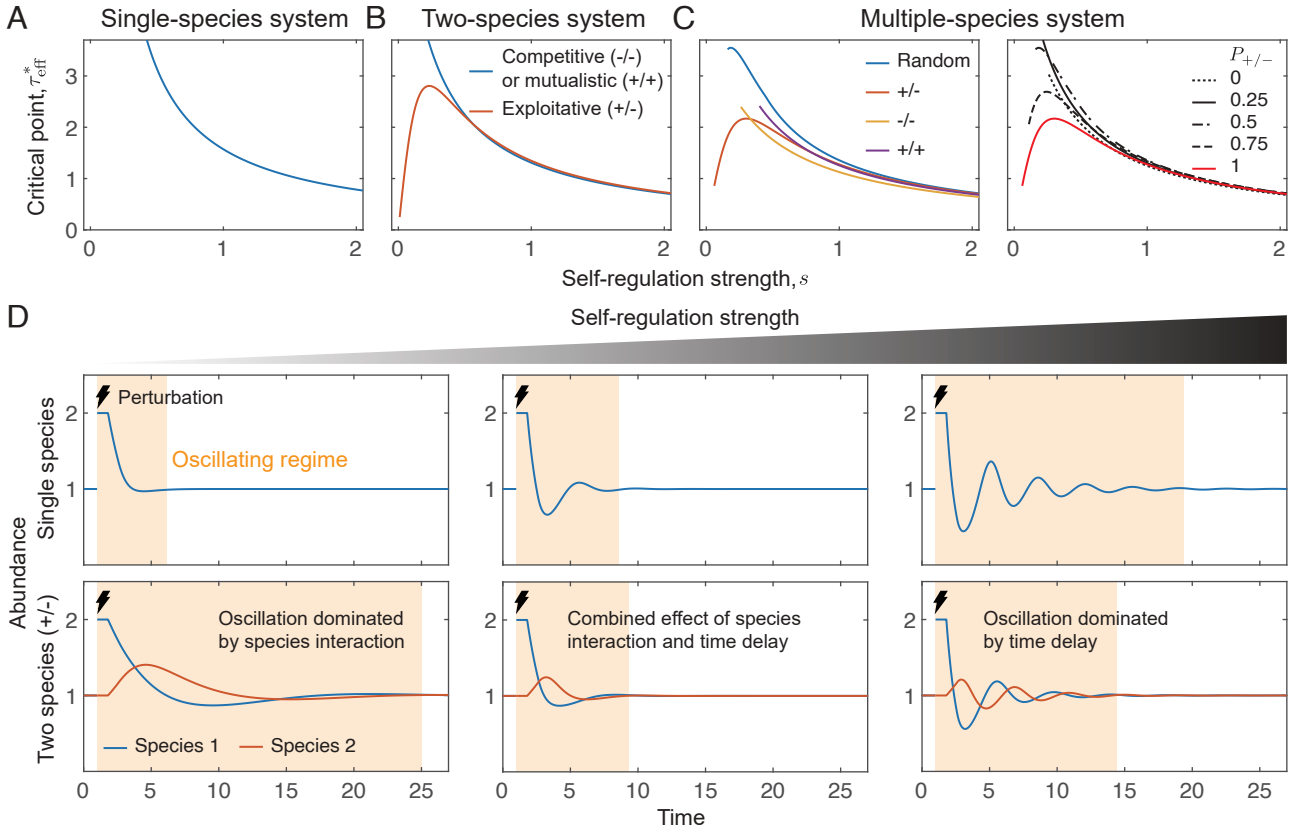


Fig. 5. Strong self-regulation promotes the occurrence of critical transitions. (*A–C*) Theoretically estimated relationships between self-regulation strength s and the critical point τ^*_{eff} for different types of systems. In the *Right* panel of (*C*), different types of lines represent different proportions of exploitative interactions ($P_{+/-}$). Competitive and mutualistic interactions occupy the same proportion: $P_{-/-} = P_{+/+} = (1 - P_{+/-})/2$. (*D*) Responses of single-species and two-species exploitative systems with different levels of self-regulation to external perturbations. Orange regions denote the oscillating regimes. Abundance oscillations can be regarded as the source of instability in time-delayed systems: when oscillations can finally be dissipated, the system can keep stability and stay away from the critical transition. For the single-species case (upper row), abundance oscillations are solely induced by time delay, and the increase of self-regulation can enhance such delay-induced oscillations. Therefore, abundance oscillations are monotonically enhanced by increasing self-regulation strength, resulting in the monotonic $\tau^*_{\text{eff}} - s$ relationship shown in (*A*). For the two-species exploitative case (bottom row), abundance oscillations are composed of two parts: exploitative interaction-induced oscillations and delay-induced oscillations. The increase of self-regulation can reduce the oscillations induced by exploitative interaction but enhance the delay-induced oscillations. When self-regulation is weak, the abundance oscillations are dominated by the exploitative interaction. Therefore, abundance oscillations are first diminished (weakening of exploitative interaction-induced oscillation) and then amplified (via the delay-induced oscillation) with increasing self-regulation strength. This results in the nonmonotonic $\tau^*_{\text{eff}} - s$ relationship of exploitative systems shown in (*B*). Altogether, the results show that strong self-regulation facilitates the occurrence of critical transitions, thereby acting as a catalyst for such regime shifts. In (*B*), interaction strengths are 0.2 (i.e., $|J_{12}| = |J_{21}| = 0.2$). In (*C*), $S = 100$, $C = 0.1$, and $\sigma = 0.05$. Numerical simulations in (*D*) are based on the generalized Lotka–Volterra model. For single-species cases, we set $s = 0.5, 1, 1.5$ for the three cases respectively. For two-species cases, we set $A_{12} = -0.2$, $A_{21} = 0.2$, and $s = 0.2, 0.6$, and 1.2 for the three cases respectively.

Although initially counterintuitive, this phenomenon can be explained by the oscillatory nature of the dynamics. Previous work has shown that critical transitions in time-delayed systems arise when external perturbations are amplified into sustained oscillations (36, 39). Relative to competitive and mutualistic interactions, exploitative interactions are inherently more oscillatory. As a result, the dominant source of oscillations differs depending on their prevalence.

When exploitative interactions are rare, oscillations are driven mainly by the time delay. In this case, stronger self-regulation enhances the delay-driven oscillations (Fig. 5*D*), thereby lowering the critical point and bringing the system closer to a transition. By contrast, when exploitative interactions are abundant, both delay and interaction structure contribute to oscillations. At low levels of self-regulation, the oscillatory tendency is dominated by exploitative interactions, so increasing self-regulation suppresses overall fluctuations and moves the system away from the critical point (Fig. 5*D*). At higher levels of self-regulation, however, delay-driven oscillations prevail, and further increases in self-regulation once again amplify fluctuations, pushing the system closer to the critical point (Fig. 5*D*).

In addition to identifying the relationship between the critical point and self-regulation across different system types, we uncover another notable result: high self-regulation strength consistently corresponds to lower critical points (Fig. 5 *A–C*). Beyond model systems, this pattern is also observed in empirical microbial systems, where low critical points are associated with strong self-regulation (Fig. 6 and *SI Appendix*, Fig. S18). This counterintuitive outcome can be explained from the perspective of oscillations. As noted above, time delays create a mismatch between instantaneous species abundance and the corresponding growth rate, generating abundance oscillations. Once delay or equilibrium abundance exceeds a threshold, these oscillations can no longer be dissipated, inducing a critical transition. Strong self-regulation intensifies species' responses to deviations from equilibrium. In delay-free systems, such responses accelerate recovery. With delay, however, species react to outdated states, and the intensified responses worsen the mismatch between abundance and growth rate. Consequently, the critical delay or equilibrium abundance at which oscillations can no longer be dissipated is reduced under strong self-regulation, thereby lowering the critical point. Taken together, this result indicates

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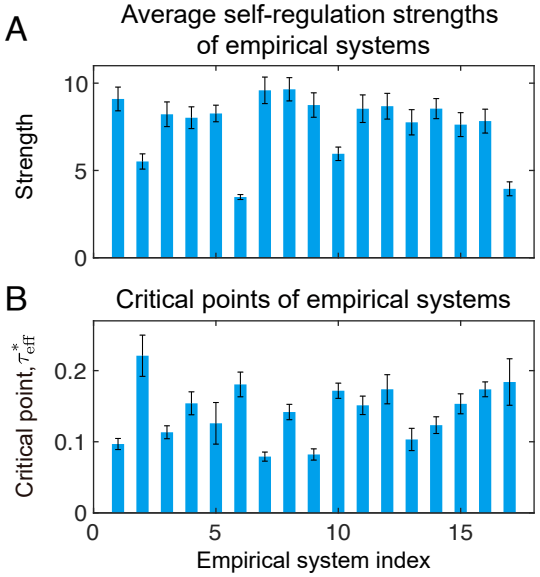


Fig. 6. Lower critical points are associated with stronger self-regulation strengths in empirical microbial systems. (*A*) Self-regulation strengths and (*B*) critical points for 17 empirical soil microbial systems (53). Bars indicate means, and error bars represent SD. Correlation analysis shows a negative association between self-regulation strength and critical point (Pearson's $r = -0.6868$, $P = 0.0023$). These results show that stronger self-regulation strengths are associated with lower critical points, supporting the prediction that strong self-regulation acts as a catalyst for critical transitions.

that—contrary to the prevailing understanding for delay-free systems, where stronger self-regulation generally promotes stability—in the presence of delays, increasing self-regulation can instead push a system toward, and even beyond, the critical point. This insight suggests that conservation efforts should carefully consider ecosystems characterized by strong self-regulation, as such systems may be closer to critical transitions and thus face elevated risks of widespread extinctions.

Discussion

Delays in species interactions are expected across ecological systems of many kinds, arising from processes that influence how organisms respond to changes in their environment or in other species (30–39). For instance, developmental lags associated with life stages can delay the response of populations to shifts in resource availability or density, because individuals must reach reproductive maturity before contributing to growth (30, 35). Such delays have long been recognized as influential in shaping ecological dynamics, and they are expected to affect when and how critical transitions occur. Yet the mathematical challenges of analyzing time-delayed, species-rich systems have limited a systematic understanding of these effects.

To address this gap, we develop a framework based on time-delayed dynamics to investigate the catalysts and inhibitors of critical transitions. Supported by analysis of empirical microbial data, our results indicate that some established principles of ecological stability require revision once delays are taken into account. Specifically, we find that a diversity of interaction types can inhibit the emergence of critical transitions, while strong self-regulation can act as a catalyst for such transitions. Our framework extends the classical random matrix approach widely used to study coexistence, stability, and reactivity in ecological networks (16–22, 25, 26, 38, 39, 47, 54, 55). Importantly, when the time delay parameter is set to zero, the model reduces to the classic random matrix framework, thereby recovering established results.

An essential component of studies on critical transitions is the development and application of early warning signals (3–5, 10, 27–29). Based on the phenomenon of critical slowing down—whereby a system recovers more slowly from perturbations as it approaches a critical transition—several indicators, including rising variance, increasing autocorrelation, and declining recovery rates, have been proposed and empirically validated in microbial experiments. In particular, these indicators were observed in cyanobacterial and yeast populations prior to fold bifurcations (28, 29), establishing that critical slowing down can be detected experimentally and used as a practical warning tool. The present study focuses on Hopf bifurcations, which, like fold bifurcations, are also accompanied by critical slowing down (3, 39). Therefore, the same early warning signals established in previous experimental work are, in principle, applicable to the oscillatory critical transitions examined here.

Our analysis has concentrated on ecological systems that settle near feasible and locally stable equilibria, as many natural communities are observed to maintain such steady states (56). Nonetheless, numerous theoretical and empirical studies have documented nonequilibrium behaviors, particularly when spatial processes are involved. Predator–prey systems, for example, can generate recurrent population cycles (35, 57), and spatial interactions can give rise to self-organized Turing patterns (58, 59). Moreover, predator–prey models have the potential to undergo period-doubling bifurcations that lead to chaotic dynamics (60), and recent empirical evidence suggests that such chaotic fluctuations may be relatively common in nature (61). Extending our framework to include delay-associated critical points in systems with inherently nonequilibrium dynamics represents an important direction for future work. We note, however, that by focusing on equilibria, we necessarily restrict attention to local bifurcations—situations where small parameter changes alter the stability of a single equilibrium state—rather than global bifurcations that involve large-scale reorganizations of system dynamics.

Indeed, changes in environmental conditions can also give rise to global bifurcations (62). These occur when larger invariant sets of the system—such as cycles or chaotic attractors—intersect or collide in phase space, often in association with nonequilibrium dynamics (46). Prior work on exploitative systems has shown that certain predator–prey models can undergo such global bifurcations (62). Extending the present framework to study delay-associated global bifurcations in complex systems is therefore another important but challenging direction for future research, as these events are often difficult to detect. More broadly, our results provide a foundation for disentangling the drivers of critical points in complex living systems and for advancing the understanding of when and how abrupt ecological transitions may occur.

Materials and Methods

Constructing Interaction Matrices. To model species-rich systems with different types of interactions, we construct the interaction matrix **J** using a random variable Z with mean 0 and variance σ^2. For each pair of species i and j, an interaction occurs with probability C, and the corresponding entries J_{ij} and J_{ji} are assigned as follows:

- With probability $P_{+/+}$, the pair interacts mutualistically, and both J_{ij} and J_{ji} are independently drawn from $|Z|$.
- With probability $P_{-/-}$, the pair interacts competitively, and both J_{ij} and J_{ji} are independently drawn from $-|Z|$.

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- With probability $P_{+/-}$, the pair interacts exploitatively, with J_{ij} and J_{ji} taking opposite signs, one drawn from $|Z|$ and the other from $-|Z|$.

The diagonal terms are set to $J_{ii} = -s$, representing self-regulation. The special cases $P_{+/+} = 1$, $P_{-/-} = 1$, and $P_{+/-} = 1$ recover purely mutualistic, competitive, and exploitative systems, respectively.

From this construction, the statistical properties of $\mathbf{J}$ for mixed-interaction systems are given by

$$\begin{cases} \mathrm{E}\left(J_{ij}\right)_{i\neq j} &= C\,\mathrm{E}(|Z|)\ (P_{+/+} - P_{-/-}), \\ \mathrm{Var}\left(J_{ij}\right)_{i\neq j} &= C\sigma^2 - \left(\mathrm{E}\left(J_{ij}\right)_{i\neq j}\right)^2, \\ \mathrm{E}\left(J_{ij}J_{ji}\right)_{i\neq j} &= C\,\mathrm{E}^2(|Z|)\ (P_{+/+} + P_{-/-} - P_{+/-}). \end{cases} \quad [13]$$

Estimating the Critical Point of Systems with Mixed-Interaction Types. For notational convenience, we define $\mathrm{E}\left(J_{ij}\right)_{i\neq j} = E$, $\mathrm{Var}\left(J_{ij}\right)_{i\neq j} = V$, and $\mathrm{E}\left(J_{ij}J_{ji}\right)_{i\neq j} = \rho$.

Random matrix theory and results on low-rank perturbations (17, 19, 25, 48, 49) describe how the eigenvalues of the interaction matrix $\mathbf{J}$ are distributed.

- Case 1: $|E| \le \sqrt{V/S}$. In this regime, all eigenvalues of $\mathbf{J}$ lie within an ellipse centered at $(-s - E, 0)$, with horizontal radius

$$r_x = \sqrt{SV}\left(1 + \frac{\rho - E^2}{V}\right), \quad [14]$$

and vertical radius

$$r_y = \sqrt{SV}\left(1 - \frac{\rho - E^2}{V}\right). \quad [15]$$

The relevant extremal points of this distribution are

$$\begin{cases} Q_{\text{leftmost}} = (-s - E - r_x, 0), \\ Q_{\text{uppermost}} = (-s - E, r_y). \end{cases} \quad [16]$$

- Case 2: $|E| > \sqrt{V/S}$. In this case, all but one eigenvalue remain within the same ellipse, while an additional outlier appears at

$$\lambda_{\text{outlier}} = -s + (S-1)E. \quad [17]$$

The extremal points are then

$$Q_{\text{leftmost}} = \left(\min\left(-s - E - r_x,\ -s + (S-1)E\right), 0\right) \quad [18]$$

and

$$Q_{\text{uppermost}} = (-s - E,\ r_y). \quad [19]$$

With these eigenvalue distributions characterized, we can estimate the critical point.

- For $|E| \le \sqrt{V/S}$,

$$\tau^*_{\text{eff}} = \min\left(\frac{\pi}{2\,(s + E + r_x)},\ \frac{\pi}{2\sqrt{(s+E)^2 + r_y^2}} - \frac{\tan^{-1}\left(\frac{r_y}{s+E}\right)}{\sqrt{(s+E)^2 + r_y^2}}\right). \quad [20]$$

- For $|E| > \sqrt{V/S}$,

$$\tau^*_{\text{eff}} = \min\left(\frac{\pi}{2\,(s + E + r_x)},\ \frac{\pi}{2\,(s - (S-1)E)},\ \frac{\pi}{2\sqrt{(s+E)^2 + r_y^2}} - \frac{\tan^{-1}\left(\frac{r_y}{s+E}\right)}{\sqrt{(s+E)^2 + r_y^2}}\right). \quad [21]$$

Robustness Test of Theoretical Results in Systems with Heterogeneous Delays and Equilibrium Species Abundances. In our main analyses, we assumed that time delays and equilibrium abundances were homogeneous across species–that is, all interactions shared the same delay τ, and all species had identical equilibrium abundances X^*. To evaluate whether our theoretical results hold when these assumptions are relaxed, we performed numerical simulations using Lotka–Volterra systems with S species:

$$\frac{\mathrm{d}X_i(t)}{\mathrm{d}t} = X_i(t)\left(r_i + \sum_{j=1}^{S} A_{ij}\,X_j(t - \tau_{ij})\right). \quad [22]$$

Here, $X_i(t)$ is the abundance of species i at time t, r_i its intrinsic growth rate, s_i the strength of self-regulation, A_{ij} the effect of species j on species i, and τ_{ij} the corresponding interaction delay. The equilibrium vector $\mathbf{X}^*$ satisfies $\mathbf{r} + \mathbf{A}\mathbf{X}^* = \mathbf{0}$. For these simulations, the interaction matrix $\mathbf{A}$ was generated as described above, delays τ_{ij} were sampled from a distribution with mean μ_τ and variance σ^2_τ, and equilibrium abundances X^*_i were sampled from a distribution with mean μ_X and variance σ^2_X (with r_i set accordingly).

In this heterogeneous setting, the effective delay defined in the main text is replaced by an average effective delay,

$$\bar{\tau}_{\text{eff}} = \mu_\tau \mu_X. \quad [23]$$

We then implemented the robustness test through the following procedure:

1. Select the type of focal system (random, exploitative, competitive, or mutualistic).
2. For a fixed value of $\bar{\tau}_{\text{eff}}$, generate N_t Lotka–Volterra systems with the same set of system parameters.
3. Apply perturbations to each of the N_t systems and record the proportion that recover to equilibrium.
4. Increment $\bar{\tau}_{\text{eff}}$ and repeat Steps 2 to 3.

Critical Point Analysis of Empirical Microbial Systems. In this study, we analyzed the critical points of 17 empirical microbial systems (53). Interaction coefficients, intrinsic growth rates, and carrying capacities were previously inferred within the generalized Lotka–Volterra framework. Using these parameters, we numerically simulated system dynamics under different time delays to test whether each system could recover from external perturbations. The critical delay was defined as the threshold beyond which recovery failed (i.e., the occurrence of a critical transition), and the corresponding critical point was calculated by multiplying this delay by the average equilibrium species abundance. Further details are provided in *SI Appendix*, section 4 and Figs. S16–S18.

Data, Materials, and Software Availability. Simulation data and analysis code have been deposited in GitHub (https://github.com/ygyang-research/CriticalTransitionDelayedDynamics2025) (63). Previously published data were used for this work (53).

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